Data Structures - Internal Test 1

Name: _____ (de-identified) Roll No: S-101

Max Marks: 10 Time: 30 min

Q1. Define a stack and name its main operations.

A stack is a linear data structure that follows the LIFO (Last In, First Out) principle, so the element inserted last is the first one removed.

Its two main operations are push, which inserts an element on the top, and pop, which removes the top element.

Q2. Explain how binary search works.

Binary search finds an element in a SORTED array, so the array must be sorted before we can apply it.

We compare the target with the middle element. If it is equal we are done. If the target is smaller we search the left half, otherwise the right half. Each step halves the search space.

The time complexity is O(log n) because the array is halved every iteration.